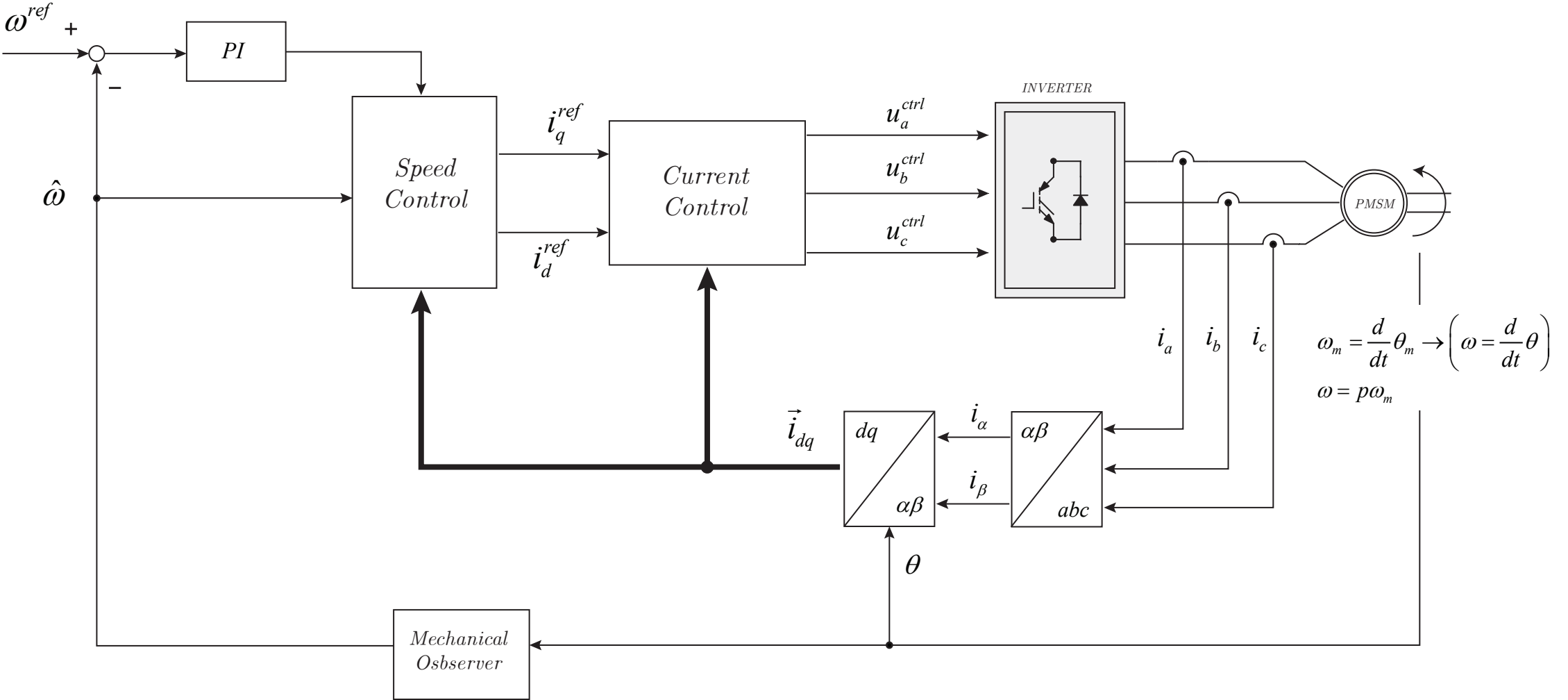

$\omega^{ref}$
+
−
PI
$\hat{\omega}$
Speed Control
$i_q^{ref}$
$i_d^{ref}$
Current Control
$u_a^{ctrl}$
$u_b^{ctrl}$
$u_c^{ctrl}$
INVERTER
PMSM
$i_a$
$i_b$
$i_c$
$\omega_m = \frac{d}{dt}\theta_m \rightarrow \left(\omega = \frac{d}{dt}\theta\right)$
$\omega = p\omega_m$
$\vec{i}_{dq}$
dq
αβ
$i_\alpha$
$i_\beta$
αβ
abc
$\theta$
Mechanical Osbserver